Engineering Design Process Activities

Spaghetti Tower Challenge

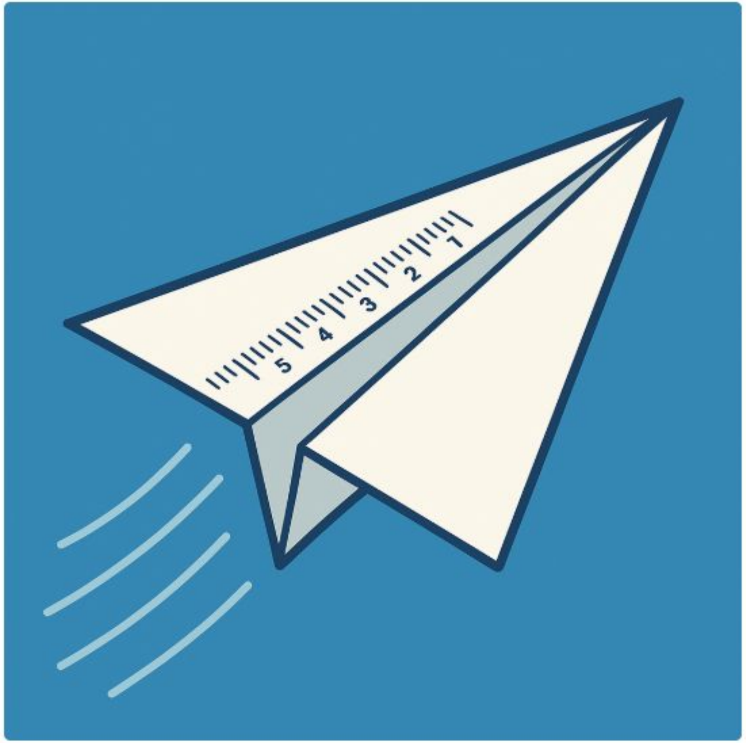
Team Name / Date: ____________________

1. **Height Achieved:** _______ inches (or cm)
2. **What Worked Well:**
   -
3. **Biggest Challenge:**
   -
4. **One Change for Next Time:**
   -
5. **Key Takeaway:**
   -

Paper Glider Challenge

Objectives

- Translate 2D sketches into a flying prototype
- Explore basic aerodynamics: lift, drag, stability
- Practice data-driven iteration



Paper Glider Challenge

Materials

- 2 sheets of 8½″×11″ paper per team
- Ruler or measuring tape
- Stopwatch or timer
- Flight zone marked on the floor (tape or mats)

Paper Glider Challenge

Procedure

1. **Sketch:** Draw your glider plan.

2. **Build:** Fold and assemble.

3. **Test Flights:**
    - Launch from the same line, three trials per team
    - Measure and record each flight distance

Team Name: ___________________

Date: _________________________

1. What Worked Well

2. Biggest Challenge

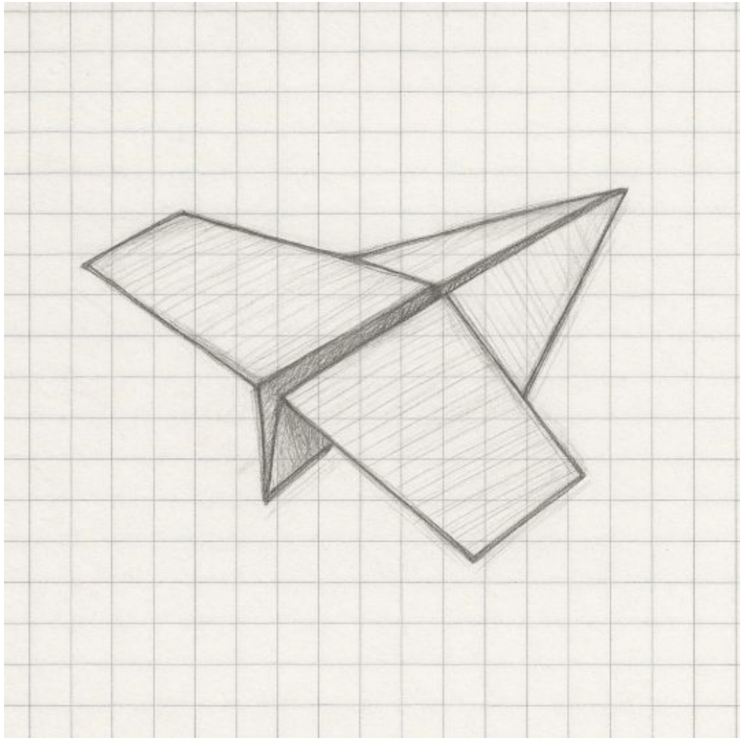
3. One Change for Next Time

4. Key Aerodynamic Principle Observed

(e.g. “Adding dihedral increased roll stability.”)

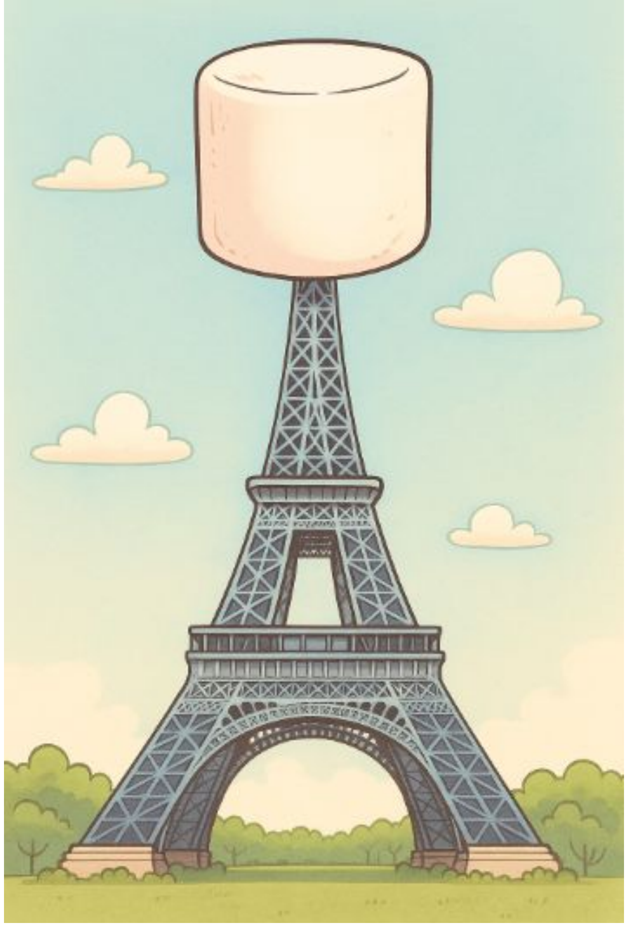
5. Sketch or describe your design

6. Which steps of the Engineering Design Process did you practice today?



Spaghetti Tower Challenge

The **rules are easy**; in 18 minutes, each group can use 20 sticks of spaghetti, one yard of masking tape, one yard of string, and one marshmallow to build the tallest free-standing structure with the entire marshmallow on the top.



For personal reflection:

- What was the strategy of the team?
- What was my role in the team?
- What worked well in my team?
- What would I improve next time in my team’s work?

For sharing within the team:

- What worked well in my team?
- What roles emerged?
- Is there work to be done on team roles?

Name:

Date:

1. **Height Achieved:** _______ inches (or cm)
2. **What Worked Well:**
3. **Biggest Challenge:**
4. **One Thing You Would Do Different:**
5. **Which steps of the Engineering Design Process did you practice today?**